

ANNOTATED CODE DOCUMENT

Project: Transportation Systems

Client: Federal Highway Administration (FHWA)

Python Code	Annotations
<pre>import pandas as pd</pre>	Imports the pandas library with the alias pd. Pandas is the main tool for reading CSV files and working with tabular data using DataFrames.
<pre>import matplotlib.pyplot as plt</pre>	Imports matplotlib's pyplot module with the alias plt. This library is used to create clear line plots for time-series visualization.
<pre>nhts_df = pd.read_csv('NHTS.csv')</pre>	Loads the National Household Travel Survey (NHTS) dataset from a CSV file into a pandas DataFrame called nhts_df. This dataset is loaded but not used in the current analysis.
<pre>ngsim_df = pd.read_csv('NGSIM.csv')</pre>	Loads the NGSIM (Next Generation Simulation) vehicle trajectory dataset into a pandas DataFrame called ngsim_df. This is the main dataset for studying car-following behavior on highways.
<pre>single_car = ngsim_df[ngsim_df['trajectory_number'] == 1].sort_values(by='Time')</pre>	Filters the NGSIM data to keep only records belonging to vehicle Trajectory #1 , then sorts the data chronologically by the Time column. This creates a clean time-series subset for one specific vehicle and its leader.
<pre>plt.figure(figsize=(10, 5))</pre>	Creates a new plotting figure with a width of 10 inches and height of 5 inches for a good aspect ratio.
<pre>plt.plot(single_car['Time'], single_car['leader_speed(m/s)'], color='blue', label='Lead Vehicle')</pre>	Plots the leader vehicle's speed (in m/s) over time using a blue line. The label parameter prepares the entry for the legend.

<code>plt.title('Time-Series: Leader Speed (Vehicle Trajectory #1)')</code>	Adds a descriptive title to the plot so the reader immediately understands what is shown.
<code>plt.xlabel('Time (seconds)')</code>	Labels the x-axis to indicate that time is measured in seconds.
<code>plt.ylabel('Speed (m/s)')</code>	Labels the y-axis to show that vehicle speed is measured in meters per second.
<code>plt.grid(True, linestyle='--', alpha=0.7)</code>	Adds a light dashed grid in the background to make the plot easier to read.
<code>plt.legend()</code>	Displays the legend on the plot (e.g., “Lead Vehicle”).
<code>plt.show()</code>	Renders and displays the completed leader speed plot.
<code>plt.plot(single_car['Time'], single_car['follower_speed(m/s)'], color='green', label='Following Vehicle')</code>	Plots the following vehicle’s speed over time using a green line. This helps visualize how the follower reacts to the leader’s speed changes.
<code>plt.title('Time-Series: Follower Speed (Vehicle Trajectory #1)')</code>	Sets a clear title for the follower speed plot.
<code>plt.xlabel('Time (seconds)') plt.ylabel('Speed (m/s)') plt.grid(True, linestyle='--', alpha=0.7) plt.legend() plt.show()</code>	Repeats the axis labels, grid, legend, and display commands for the follower speed plot.